

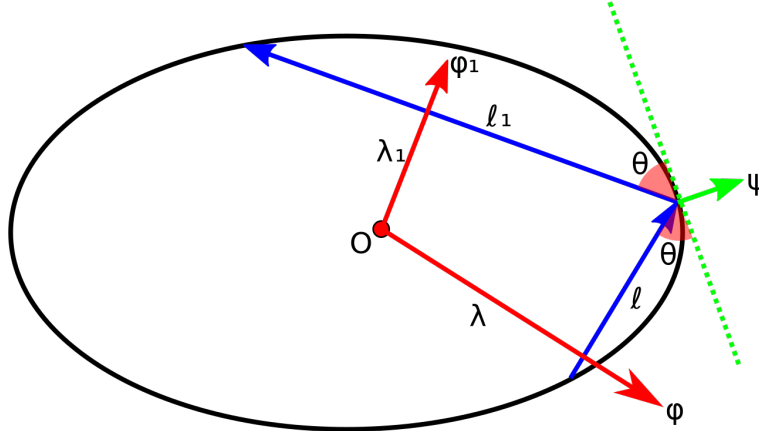


FIGURE 1. The billiard map $f(\varphi, \lambda) = (\varphi_1, \lambda_1)$, the normal angle $\psi = (\varphi_1 + \varphi)/2$ and the incidence-reflection angle $\theta = (\varphi_1 - \varphi)/2$.

Let $p/q \in (0, 1)$ be a rational number with $\gcd(p, q) = 1$. We call the elements of the space

$$X = \{ \{ \varphi_j \} \in \mathbb{R}^{\mathbb{Z}} : \varphi_j < \varphi_{j+1} < \varphi_j + 2\pi, \varphi_{j+q} = \varphi_j + 2\pi p, \forall j \in \mathbb{Z} \}$$

p/q -periodic sequences. We define the p/q -periodic action $A : X \rightarrow \mathbb{R}$ as

$$A\{\varphi_j\} = \sum_{j=0}^{q-1} S(\varphi_j, \varphi_{j+1}) = 2 \sum_{j=0}^{q-1} h(\psi_j) \sin \theta_j,$$

where $\psi_j = \sigma\{\varphi_j\}/2$ and $\theta_j = \delta\{\varphi_j\}/2$. Periodicities $\varphi_{j+q} = \varphi_j + 2\pi p$, $\psi_{j+q} = \psi_j + 2\pi p$ and $\theta_{j+q} = \theta_j$ imply that $\mu\{S(\varphi_{j_0+j}, \varphi_{j_0+j+1})\} = 2\mu\{h(\psi_{j_0+j}) \sin \theta_{j_0+j}\} = A\{\varphi_j\}/q$ for all $j_0 \in \mathbb{Z}$. Critical points of the action, which we call p/q -periodic configurations, can be lifted to full p/q -periodic orbits of the billiard map f by taking $\lambda_j = -\partial_1 S(\varphi_j, \varphi_{j+1}) = \partial_2 S(\varphi_{j-1}, \varphi_j)$. Thus, any p/q -periodic configuration defines a p/q -periodic billiard trajectory inside Γ with side angles φ_j , normal angles $\psi_j = \sigma\{\varphi_j\}/2$ and incidence-reflection angles $\theta_j = \delta\{\varphi_j\}/2$. We characterize such configurations in the next proposition, where we also recall a formula for the length of their corresponding periodic billiard trajectories.

Proposition 3. A p/q -periodic sequence $\{\varphi_j\}$ is a p/q -periodic configuration if and only if

$$\sigma\{h'(\psi_j) \sin \theta_j\} - \delta\{h(\psi_j) \cos \theta_j\} = 0, \quad \forall j \in \mathbb{Z},$$

where $\psi_j = \sigma\{\varphi_j\}/2$ and $\theta_j = \delta\{\varphi_j\}/2$, in which case

$$2q\mu\{h(\psi_j) \sin \theta_j\} = A\{\varphi_j\} = L, \quad \mu\{h'(\psi_j) \sin \theta_j\} = 0, \quad \mu\{\theta_j\} = \pi p/q,$$

where L is the length of the p/q -periodic configuration $\{\varphi_j\}$.

Proof. A p/q -periodic sequence $\{\varphi_j\}$ is a critical point of the action if and only if

$$\sigma\{h'(\psi_j) \sin \theta_j\} - \delta\{h(\psi_j) \cos \theta_j\} = \partial_2 S(\varphi_j, \varphi_{j+1}) + \partial_1 S(\varphi_{j+1}, \varphi_{j+2}) = 0$$

for all $j \in \mathbb{Z}$. The first equality above follows from $S(\varphi_j, \varphi_{j+1}) = 2h(\psi_j) \sin \theta_j$ and relations $\psi_j = \sigma\{\varphi_j\}/2 = (\varphi_{j+1} + \varphi_j)/2$ and $\theta_j = \delta\{\varphi_j\}/2 = (\varphi_{j+1} - \varphi_j)/2$.